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$$z^4 = 8(-1 + i\sqrt{3})$$

$$r = 8\sqrt{(-1)^2 + (\sqrt{3})^2} = 16$$

$$\theta = \frac{2\pi}{3}$$

$$z_k = 2 \left(\cos \left(\frac{\frac{2\pi}{3} + 2k\pi}{4} \right) + i \sin \left(\frac{\frac{2\pi}{3} + 2k\pi}{4} \right) \right)$$

$$z_0 = \sqrt{3} + i$$

$$z_1 = -1 + i\sqrt{3}$$

$$z_2 = -\sqrt{3} - i$$

$$z_3 = 1 - i\sqrt{3}$$

References